
EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 2A

1. Span Basics

(a) Write the span $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$ as a linear combination of its basis vectors.

(b) Is $\begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}$ in $\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$?

(c) What is a possible choice for $\mathbf{v}$ that would make $\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v} \right\} = \mathbb{R}^3$?

2. Span Proof

Given some set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, show the following:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \text{span}\{\alpha\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}, \text{ where } \alpha \text{ is a non-zero scalar}$$

In other words, we can scale our spanning vectors and not change their span.

3. Exploring Dimension and Linear Independence

Let's consider the vector space $\mathbb{R}^k$ and a set of n vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{R}^k$.

- (a) For the first part of the problem, let $k > n$. Can $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span the full $\mathbb{R}^k$ space? If so, justify it. If not, what conditions does it violate/what is missing?

- (b) Let $k = n$. Can $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span the full $\mathbb{R}^k$ space? Why/why not? What conditions would we need?

- (c) Finally, let $k < n$. Can $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span the full $\mathbb{R}^k$ space?